# Assignment 3

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## 1 Introduction

The objective of this assignment is to simulate the diffusion of information through a social network where different types of actors coexist.
Three different kinds of information are considered:

- a reliable message propagated by Verifiers;
- a mutated message generated by Unintentional Spreaders;
- fake information introduced by Social Bots.

The goal is to analyze how heterogeneous node behaviors influence the diffusion process and how misinformation may propagate through the network.

## 2 Actor Roles

The simulation includes four types of nodes, each associated with a specific message behavior. Table 1 summarizes their characteristics.

| **Role** | **Behavior** | **Message** |
|---|---|---|
| Verifier | Reliable source that always broadcasts the correct information and never changes state. | Real |
| Unintentional Spreader | Accidentally alters the true message once upon adoption and forwards the modified version. | Altered |
| Normal Node | Adopts a message if enough neighbors support it (threshold rule: $\geq \theta$). | Majority or inactive |
| Social Bot | Replaces any received message with fake information and spreads it aggressively. | Fake |

Table 1: Summary of node roles and message behaviors.

## 3 Diffusion Model

The spreading process evolves in discrete iterations. At every iteration:

1. Verifiers keep broadcasting the true message.
2. Unintentional nodes may transform the true message into the mutated one.
3. Normal nodes adopt the majority message if the threshold condition is satisfied.
4. Bots replace every received message with fake information.

The simulation terminates when the network reaches a stable state. A maximum of 20 iterations is used as a safety limit.

## 4 Preliminary Validation on a Small Graph

Before applying the diffusion model to the full anime network, a preliminary experiment was conducted on a small synthetic graph in order to validate the correctness of the implementation and to observe the expected qualitative behavior of the model.
The small graph consisted of 10 nodes with random connectivity. To evaluate the sensitivity of the model, the number of Unintentional Spreaders was progressively increased, and the resulting fraction of mutated messages was recorded.

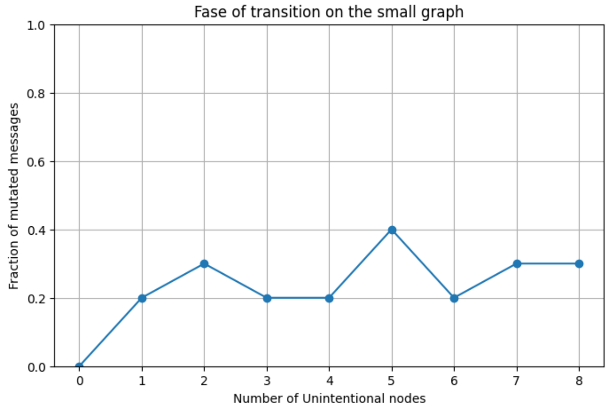


Figure 1: Fraction of mutated messages as the number of Unintentional Spreaders increases in the small graph.

With only 10 nodes, even a single mutated adoption corresponds to 10% of the network, making the curve extremely sensitive to small structural variations. Moreover, the position of Unintentional Spreaders plays a dominant role: when an Unintentional node lies on a central path, the mutated message can reach several neighbors, producing peaks such as 0.4 for five Unintentional nodes. Conversely, when Unintentional nodes are placed in peripheral regions, the mutated message fails to propagate, resulting in lower values.

This is a characteristic of small random graphs, where the threshold condition is easily satisfied and local structures strongly influence the diffusion outcome. These preliminary results confirm that the model behaves correctly and highlight the importance of network size and connectivity in stabilizing the diffusion process.

# 5 Diffusion on the Anime Network

After validating the model on a small synthetic graph, the diffusion process was applied to the full anime similarity network. To isolate the main factors influencing accidental misinformation, four controlled experiments were performed: (i) random placement of Unintentional nodes, (ii) placement on high-degree nodes, (iii) variation of the adoption threshold, (iv) variation of the number of Verifiers.

## 5.1 Random vs High-Degree Unintentional Spreaders

Figures 2 and 3 compare the effect of random placement and high-degree placement. Random Unintentional nodes produce extremely low mutated fractions (below 4.36%), since they rarely occupy structurally influential positions. Selecting Unintentional nodes among the highest-degree nodes slightly increases mutation (up to 7.18%), but even structurally central nodes fail to trigger large-scale mutation due to the dominance of Verifiers and the dense connectivity of the network.

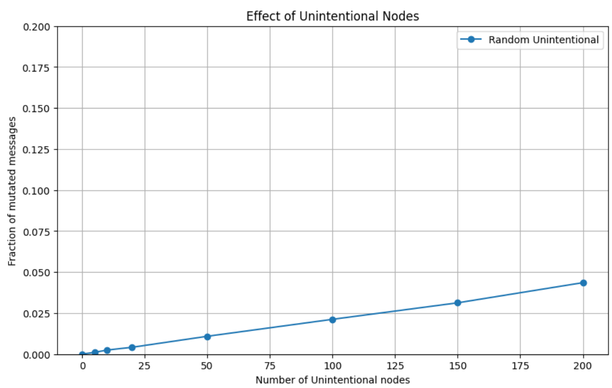


Figure 2: Random Unintentional nodes

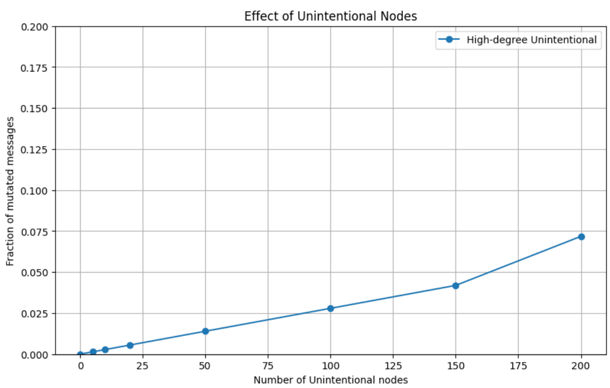


Figure 3: High-degree Unintentional nodes

## 5.2 Threshold and Verifier Effects

Figures 4 and 5 shows the impact of the adoption threshold and the number of Verifiers. The threshold sweep reveals a sharp transition: for $\theta \leq 0.05$, a small amount of mutation appears (around 1%), while for $\theta \geq 0.1$ the mutated message disappears entirely. Similarly, varying the number of Verifiers between 10 and 200 has negligible impact: the mutated fraction remains around 1% in all cases. These results indicate that the network's topology alone stabilizes the diffusion process, with Verifiers reinforcing an already robust configuration.

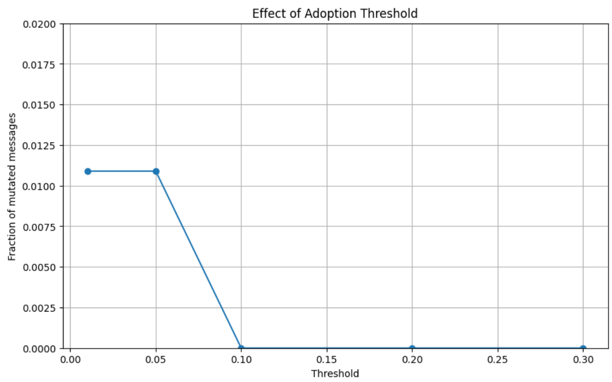


Figure 4: Threshold sweep

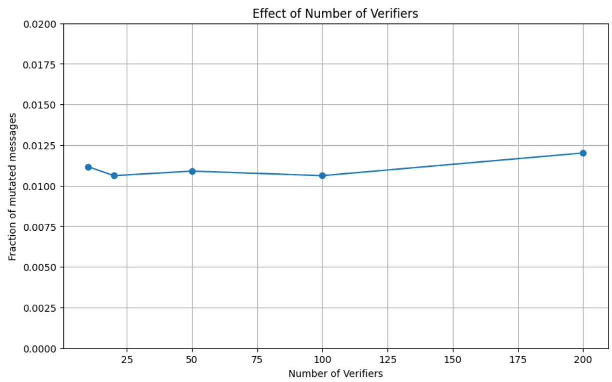


Figure 5: Verifier sweep

## 5.3 Summary

Across all experiments, the mutated message remains confined to a very small portion of the network. The anime similarity graph is structurally robust: its high average degree, dense connectivity, and distributed verifier presence ensure that the true message dominates the diffusion process under a wide range of conditions.

# 6 Phase Transition Analysis

A key question of this assignment is to determine whether the anime similarity network exhibits a *phase transition* in the diffusion of mutated information, i.e., whether a small increase in the number of Unintentional Spreaders can suddenly push the mutated message above the 50% threshold. Since random placement never produced a transition, Unintentional nodes were placed on high-degree positions.

To locate this critical point efficiently, a binary search was first used to identify the approximate region where the transition occurs. This coarse search indicated a critical value around 404 Unintentional nodes.

Then a finer linear sweep was performed in a restricted interval around this value. This refined search revealed that the actual transition occurs earlier, at approximately 382, which is the smallest number of strategically placed Unintentional nodes for which the mutated message exceeds 50% of the network.

| Threshold | Critical Unintentional Nodes |
|---|---|
| 0.01 | 404 (coarse), 382 (precise) |
| 0.05 | None |
| 0.10 | None |
| 0.20 | None |
| 0.30 | None |

Table 2: Minimum number of Unintentional nodes required for a phase transition.

A phase transition is observed only for the lowest threshold ($\theta = 0.01$). For all higher thresholds, no transition occurs: even thousands of Unintentional nodes fail to make the mutated message dominant. These results confirm the structural robustness of the anime similarity network.

# 7 Diffusion with Social Bots

In the second part of the assignment, the diffusion model is extended by introducing Social Bots, i.e., malicious actors designed to actively subvert the flow of information. Unlike Unintentional Spreaders, which generate misinformation only by mistake, Bots deliberately replace any received message with fake news and propagate it aggressively to all their neighbors. Their behavior is therefore substantially more disruptive than accidental mutation.

## 7.1 Bot Mechanics

Bots follow two strengthened behavioral rules:

- **Active subversion**: as soon as a Bot receives any message (true, mutated, or fake), it immediately replaces it with fake news and broadcasts it to all neighbors.
- **Zero-threshold activation**: a single active neighbor is sufficient to trigger fake propagation. Bots do not rely on the adoption threshold used by Normal nodes.

To maximize their disruptive potential, Bots are placed strategically. Specifically, they are selected among the nodes with highest betweenness centrality, ensuring that they occupy structurally influential positions in the network.

## 7.2 Simulation Results

The diffusion process was executed twice: once using the baseline configuration of Part 1 (without Bots), and once after introducing three strategically placed Bots. Figure 6 shows the final state of the network in both scenarios.

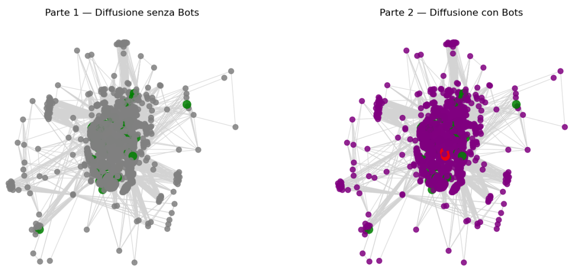


Figure 6: Comparison between diffusion without Bots (left) and with Bots (right).

| Node Role | Color |
|---|---|
| Verifier | Green |
| Unintentional Spreader | Yellow |
| Social Bot | Red |

| Message State | Color |
|---|---|
| True (`Real`) | Blue |
| Mutated (`Altered`) | Orange |
| Fake (`Fake`) | Purple |
| None | Gray |

Table 3: Color encoding used in Figure 6 for node roles (left) and message states (right).

In the left panel (diffusion without Bots), almost all nodes appear in gray, indicating that they remain inactive and do not adopt any message. Only the Verifiers (green) hold the true message, and no mutated or fake information spreads. This behavior reflects the structural robustness observed in Part 1.

In the right panel (diffusion with Bots), the network becomes dominated by purple nodes. Despite being only three and strategically placed using betweenness centrality, Bots trigger a cascade of fake-news adoption that rapidly overwhelms the network. Normal nodes adopt the fake message immediately when exposed to it, causing the fake news to spread far more aggressively than the true message.

The quantitative results are summarized below:

| Scenario | True (%) | Mutated (%) | Fake (%) | None (%) |
|---|---|---|---|---|
| Without Bots | 1.40 | 0.00 | 0.00 | 98.60 |
| With Bots | 1.40 | 0.00 | 77.82 | 20.79 |

Table 4: Final message distribution with and without Social Bots.

### 7.3 Discussion

In the absence of Bots, the anime similarity network exhibits strong resilience.
However, the introduction of only three bots completely overturns the dynamics. Fake information becomes the dominant message, reaching nearly 78% of the nodes, while both true and mutated messages collapse. Bots exploit their strategic location and zero-threshold activation to rapidly contaminate the network, demonstrating how even a small number of adversarial nodes can destabilize an otherwise stable diffusion process.

### 7.4 Summary

Social Bots represent a qualitatively different threat compared to Unintentional Spreaders. While accidental mutation remains confined and easily suppressed, malicious fake-news propagation quickly becomes dominant when Bots are introduced. This experiment illustrates the importance of node behavior and structural centrality in determining the stability of information flow within complex networks.

## 8 Final Conclusion

The results reveal a clear and meaningful distinction between accidental misinformation and intentional fake-news propagation.
The network is robust to accidental mutation but highly vulnerable to coordinated malicious behavior. The structural centrality of Bots amplifies their influence, enabling them to override the natural stability of the diffusion process.
Overall, these findings highlight how a small number of strategically positioned malicious actors can significantly compromise the reliability of information diffusion.